Lecture 8
Chapter 7: Normalized Database Design
Part 2

Md. Tariquzzaman
Junior Lecturer, Dept. of Computer Science and Engineering
Islamic University of Technology (IUT)

CSE 4307: Database Management Systems

Normal Forms

What are Normal Forms?

Normalization: Motivation (Recap)

The Goal of Relational Database Design

To generate a set of relation schemas that will meet the following two goals:

1. Allows us to **store information without unnecessary redundancy**.
2. Allows us to **retrieve information easily**.

The Evaluation Method

Hence, we need a standard method to evaluate a design, which brings us to **Normal Forms**.

Normalization: Motivation Example (Recap)

ID	Name	Salary	Dept	Build.	Budget
22222	Einstein	95000	Physics	Watson	70000
12121	Wu	90000	Finance	Painter	120000
32343	El Said	60000	History	Painter	50000
45565	Katz	75000	**Comp. Sci.**	**Taylor**	**100000**
98345	Kim	80000	Elec. Eng.	Taylor	85000
76766	Crick	72000	Biology	Watson	90000
10101	Srinivasan	65000	**Comp. Sci.**	**Taylor**	**100000**
58583	Califieri	62000	History	Painter	50000
83821	Brandt	92000	**Comp. Sci.**	**Taylor**	**100000**
15151	Mozart	40000	Music	Packard	80000
33456	Gold	87000	Physics	Watson	70000
76543	Singh	80000	Finance	Painter	120000

3 Problems:

1. **Redundancy**: Department info is repeated.
2. **Inconsistency**: Update of department info should be propagated properly.
3. **Bad Business Logic**: You cannot enter data for a new department unless there is a teacher of that department.

First Normal Form (1NF)

Atomic Domains and First Normal Form

Atomic Domain

A domain is **atomic** if elements of the domain are considered to be **indivisible units**.

- **Examples:** Height, weight, country information in general.

First Normal Form (1NF)

We say that a relation schema R is in first normal form (1NF) if:

1. The domains of all attributes of R are **atomic**.
2. And the value of each attribute contains only a **single value** from that domain.

First Normal Form: Example

ID	First Name	Family Name	Phone Number
101	a	b	**0176543, 0176548**
102	c	d	0165438
103	e	f	**0176654, 0189756**

1NF Violation

Multiple values exist in the phone number attribute (even if each individual number is considered atomic).

Apparent Solution: Add more attributes

ID	First Name	Family Name	Phone1	Phone2
101	a	b	0176543	0176548
102	c	d	0165438	NULL
103	e	f	0176654	0189756

Now it is in 1NF, but it introduces new problems:

- **Space wastage**: Many entries are null.
- **Larger design problem**: What if you want to add another phone number? And then another? The schema would need constant restructuring.

An Ideal Solution

Base Table

ID	First Name	Family Name
101	a	b
102	c	d
103	e	f

Phone Table

ID	Phone No
101	0176543
101	0176548
102	0165438
103	0176654
103	0189756

The Result

Now the schema is in 1NF. Multiple values can be added efficiently by simply inserting new rows, without altering the table structure.

1NF: Problem with Integer Values

- An integer is considered atomic **as long as it has no sub-parts**.
- Integers or strings with sub-parts create problems (e.g., $ID = CSE001$).
- Extra programming effort is needed to extract department information and serial numbers from a given ID.

The Verdict

Since we need to split the ID to get other information, the ID is no longer considered atomic. **So, the schema is NOT in 1NF.**

The Solution: If the schema contains separate attributes for each of the participating sub-parts of the ID (in this case: `dept`), then it is in 1NF. This incurs redundancy but is intentional and useful by the end-users.

Second Normal Form (2NF)

1. Definition of 2NF

The 2NF Rule

A table is in 2NF if it satisfies two conditions:

1. It is already in **1NF**.
2. It has **NO Partial Dependencies**.

What is a Partial Dependency?

This occurs when a non-key attribute depends on only *part* of a composite primary key.

$$\text{Proper Subset of PK} \longrightarrow \text{Non-Prime Attribute}$$

2. The Problem: A Partial Dependency

StudentID (PK)	CourseID (PK)	StudentName	CourseFee	Grade
101	CS101	Alice	$500	A
101	Math20	Alice	$300	B
102	CS101	Bob	$500	A

Why this violates 2NF:

- **StudentName** depends ONLY on **StudentID**.
- **CourseFee** depends ONLY on **CourseID**.

Result: Data redundancy. Alice's name and the $500 fee are needlessly repeated. Update anomalies will occur if Alice changes her name.

3. The Solution: Decomposing to 2NF

Break the table apart so every non-key column depends on the *entire* primary key.

Students

s_ID	Name
101	Alice
102	Bob

Courses

c_ID	Fee
CS101	$500
Math20	$300

Enrollments

s_ID	c_ID	Grade
101	CS101	A
101	Math20	B
102	CS101	A

Why this is 2NF

In **Enrollments**, the **Grade** attribute requires BOTH the student and the course to be identifiable. No partial dependencies remain.

2NF Mathematical Examples

2NF Test: Example 1 (Failing 2NF)

Given: $R(A, B, C, D, E, F)$
FDs: $A \to B, \quad B \to C, \quad C \to D, \quad D \to E$

Step 1: Identify Candidate Keys and Attributes

- **Candidate Key (CK):** AF *(Derived via discard process: A determines B,C,D,E. F determines itself).*
- **Proper Subsets of CK:** $\{A\}$ and $\{F\}$
- **Prime Attributes (PA):** $\{A, F\}$
- **Non-Prime Attributes (NonPA):** $R - PA = \{B, C, D, E\}$

Step 2: Test for Partial Dependency

Does any PSS$(CK) \to$ NonPA hold?

- Look at the given FD: $A \to B$
- A is a proper subset of our key AF.
- B is a Non-Prime Attribute.

Conclusion

Because part of the key (A) determines a non-key attribute (B), a **Partial Dependency exists.** R **is NOT in 2NF.**

2NF Test: Example 2 (Passing 2NF)

Given: $R(A, B, C, D)$
FDs: $AB \to CD, \quad C \to A, \quad D \to B$

Step 1: Identify Candidate Keys and Attributes

- **Candidate Keys (CKs):** $\{AB, BC, AD\}$ *(Derived via overlapping replacements).*
- **Prime Attributes (PA):** $\{A, B, C, D\}$ *(Because all attributes appear in at least one CK).*
- **Non-Prime Attributes (NonPA):** $R - PA = \{A, B, C, D\} - \{A, B, C, D\} = \emptyset$

Step 2: Test for Partial Dependency
Does any PSS$(CK) \to$ NonPA hold?

Conclusion

Since the set of Non-Prime Attributes is **empty**, it is mathematically impossible for a partial dependency to exist. There are no NonPA targets to point to.

Therefore, Relation R is inherently in 2NF.

Third Normal Form (3NF): Conditions

Definition of 3NF

A Relation R is in 3NF if the following 2 conditions are met:

1. R must be in **2NF** (i.e., No Partial Dependency, PSS of $CK_i \rightarrow$ NonPA).
2. It must **not contain any Transitivity Dependency** for Non-Prime Attributes. In other words, there must not be any FD $\alpha \rightarrow \beta$ where both α and β are **Non-Prime Attributes**.

Key Observations:

- **Note 1 (Gradual Strictness):** The rules are getting stricter.
  - In 2NF, we forbid: PA $\rightarrow$ NonPA
  - In 3NF, we forbid: NonPA $\rightarrow$ NonPA
- **Note 2 (Adequacy):** 3NF is considered adequate for normal relational database design because most 3NF tables are free of insertion, update, and deletion anomalies.

3NF Second Condition: Transitive Non-Prime Attribute Dep.

The 2nd Condition Explained

There must not be any Functional Dependency $\alpha \rightarrow \beta$ where both α and β are **Non-Prime Attributes (NonPA)**.

SID	Name	Dept	DeptEst
1	a	CSE	1998
2	b	EE	1997
3	a	CSE	1998

Here, `SID` is the **only CK**.
We observe the FDs:

- $SID \rightarrow Dept$
- $Dept \rightarrow DeptEst$

Transitive Rule implies: $SID \rightarrow DeptEst$

- Here, both `Dept` and `DeptEst` are **NonPA**.
- If we get any $X \rightarrow \alpha$ and $\alpha \rightarrow \beta$ (where X is the CK), it creates a situation where $\alpha \rightarrow \beta$ has NonPA on **both sides**.
- **This introduces redundancy!**

3NF: Example 1

Given: $R(A, B, C, D)$
FDs: $A \to B, B \to C, C \to D$

Solution Step 1: Find all CKs

(CKs are needed to find PA, which is used to find NonPA)

- Start with all: $ABCD^+ = \{A, B, C, D\}$
- **Discard:** Use Transitive Rule ($A \to B \to C \to D$)
- $\{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}\}^+ \Rightarrow A^+ = \{A, B, C, D\}$

Is A a Candidate Key?

- A is a Superkey (SK).
- Because there is no proper subset of A (it is a single attribute), **A is a CK.**

3NF: Example 1 (Cont.)

Next: Is there any more CK?

Up to now, we got $A = CK$.

- PA $= \{A\}$
- DA $= \{B, C, D\}$ *(From RHS of FDs)*

Check Intersection:

$$PA \cap DA = \emptyset$$

Conclusion for CKs

Since the intersection is empty, **no other CK is possible.**
We only have one CK: A.

3NF: Example 1 (Cont.)

Check against the 3NF condition:

$CK_1 = A$
$PA = \{A\}$
$NonPA = \{B, C, D\}$

Now check each FD ($\alpha \rightarrow \beta$) for NonPA components:

- $A \rightarrow B$: A is a PA. No need to check RHS.
- $B \rightarrow C$: Both B and C are NonPA!
- $C \rightarrow D$: Both C and D are NonPA!

Final Result

Because dependencies exist where both sides are Non-Prime Attributes, **R is NOT in 3NF. (Done)**

3NF: Example 2

Given: $R(A, B, C, D, E, F)$
FDs: $AB \rightarrow CDEF, BD \rightarrow F$

Solution Step 1: Find all CKs

- Start with all: $ABCDEF^+ = \{all\}$
- **Discard:** Use $AB \rightarrow CDEF$
- $\{A, B, \mathbf{C}, \mathbf{D}, \mathbf{E}, \mathbf{F}\}^+ \Rightarrow AB^+ = \{all\}$

Proper Subsets (PSS) of AB:

- $A^+ = \{A\}$ **X SK**
- $B^+ = \{B\}$ **X SK**

So, AB is a CK.

3NF: Example 2 (Cont.)

Step 1.2: Any other CK?

- $CK = AB \Rightarrow PA = \{A, B\}$
- $DA = \{C, D, E, F\}$ *(Right side of FDs)*
- $PA \cap DA = \emptyset \Rightarrow$ **No other CK possible.**

Now, apply 2nd Condition:

$NonPA = \{C, D, E, F\}$

Check each FD for NonPA $\rightarrow$ NonPA:

- $AB \rightarrow CDEF$: AB is a PA.
- $BD \rightarrow F$: Here, since D is a Non-Prime, BD **as a whole is also Non-Prime**. F is also Non-Prime.

Final Result

(i.e., $AB \rightarrow BD$ and $BD \rightarrow F \Rightarrow AB \rightarrow F$, which is Transitivity NonPA).

So, R is NOT in 3NF. (Done)

Boyce-Codd Normal Form (BCNF)

Boyce-Codd Normal Form (BCNF): Motivation

What is BCNF?

BCNF is the advanced version of 3NF. It is **stricter than 3NF** and is also commonly referred to as **3.5 NF**.

Historical Context:

- Developed in 1974 by **Raymond F. Boyce** and **Edgar F. Codd**.
- Designed to address certain types of anomalies not dealt with by 3NF as originally defined.

The Root Cause of 3NF Anomalies

Anomalies occur in 3NF specifically when a relation has **Multiple Overlapping Candidate Keys**.

Boyce-Codd Normal Form: Conditions

Definition of BCNF

A Relation R is in BCNF if the following 2 conditions are met:

1. R must be in **3NF** (i.e., No Transitivity Dependency on NonPA on both sides).
2. For each **Non-trivial Functional Dependency** $\alpha \rightarrow \beta$, α must be a **Superkey (SK)**.

Note on Verification:

- It is relatively easy to check. For a smaller relation R, you can readily check the Left-Hand Side (α) to see if it is a Superkey or not.
- However, for rigorous proof, we will use our **conventional process** (Method 3 closures) that we used earlier.

BCNF By Example

BCNF By Example

BCNF Example 1

Given: R(A, B, C)

FDs: $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow A$

Task: To test if R is in BCNF.

Solution Note: BCNF removed the problem for **multiple overlapping Candidate Keys** (such as AB, BC, CD).

Review Conditions:

1. Must be in **3NF**.
2. For all non-trivial dependencies $X \rightarrow Y$, **X must be a Superkey (SK)**.

Note on Methodology

We will use the same Method (3) to get all CKs. It gives other info needed for BCNF (i.e., verifying SKs).

BCNF By Example (Cont.)

Step 1: Find CKs using Method 3

- Start with all: $ABC^+ = \{A, B, C\}$
- **Discard:** Using $A \rightarrow B$ and $C \rightarrow A$, we can discard A and B.
- $C^+ = \{A, B, C\}$ **is a SK and CK**
  - *(i.e., No need to check as no Proper Subset exists)*
- **Again:** Start with $ABC^+ = \{A, B, C\}$
- **Discard:** Using $B \rightarrow C$ and $A \rightarrow B$, we can discard B and C.
- $A^+ = \{A, B, C\}$ **is a SK and CK**

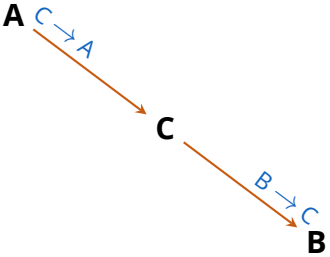
Step 2: Setup for Replacement Process

- $PA = \{A, C\}$
- $DA = \{A, B, C\}$ *(Right side of FDs)*
- $\alpha = PA \cap DA = \{A, C\}$

So far, $CK_1 = C$, $CK_2 = A$.

BCNF By Example (Cont.)

Replacement Process (RP) for A and C:



Evaluate the Replacements:

- **From A:** Replaced by C (already done).
- **From C:** Replaced by B (New).
  - B^+ = {A, B, C}. It has no PSS, **so B is a SK and CK.**

Final Check for BCNF

So, CKs are {A, B, C}. They are also SKs.

Now check the LHS of our FDs: {A, B, C}. Since all of them are SKs, **it is in BCNF.**

Real World Examples

2NF: Real Example

2NF Condition:

- No such dependency:
- **PSS(CK_i) $\rightarrow$ NonPA**
- Because such FD introduces Redundancy.

StudentCourse Table

SID	CID	Credit
1	CSE101	3
2	CSE102	4
1	CSE103	3
3	CSE101	3
2	CSE101	3

Analysis:

- Here, CK = (SID, CID) *(composite)*
- Now, we observe: **CID $\rightarrow$ Credit**
- And CID is a PSS of CK. **(Violation of 2NF)**

Right Decomposition:

- **StudentsCourse** (SID, CID(FK))
- **Courses** (CID, Credit)

3NF: Real Example

3NF Condition:

- No such Transitivity Dependency (TD):
- **NonPA $\longrightarrow$ NonPA**

Students Table

SID	Dept	DeptBudget
1	CSE	110
2	EEE	120
3	CSE	110

Analysis:

- Here we observe CK = SID
- And **Dept $\longrightarrow$ DeptBudget**
- Therefore, NonPA = {Dept, DeptBudget}. Both are NonPA.

Explanation (Transitive Dependency)

SID $\longrightarrow$ Dept and Dept $\longrightarrow$ DeptBudget $\Rightarrow$ SID $\longrightarrow$ DeptBudget
It means if you have SID and Dept then you can deduce DeptBudget, so DeptBudget in this Relation is **Redundant**.

BCNF: Real Example

Scenario: Suppose students can take courses. One course may be conducted by multiple teachers, **But one teacher can take only one course.**

BCNF Conditions:

- For each Non Trivial FD $\alpha \rightarrow \beta$
- α **is a SK**

Students Table

SID	CID	TID
1	Java	T1
1	C	T2
2	DB	T3
3	Java	T4
4	DB	T5
4	DB	T3
5	C	T2

BCNF: Real Example (Decomposition)

Analysis:

- Here, CK = (SID, CID)
- Observe the FD: **TID $\longrightarrow$ CID**
- But TID is **not a SK**. So, it is not in BCNF.

So, we decompose:

Table 1

SID	TID
1	T1
1	T2
2	T3
3	T4
4	T5
4	T3
5	T2

Table 2

TID	CID
T3	DB
T1	Java
T2	C
T4	Java

Questions?